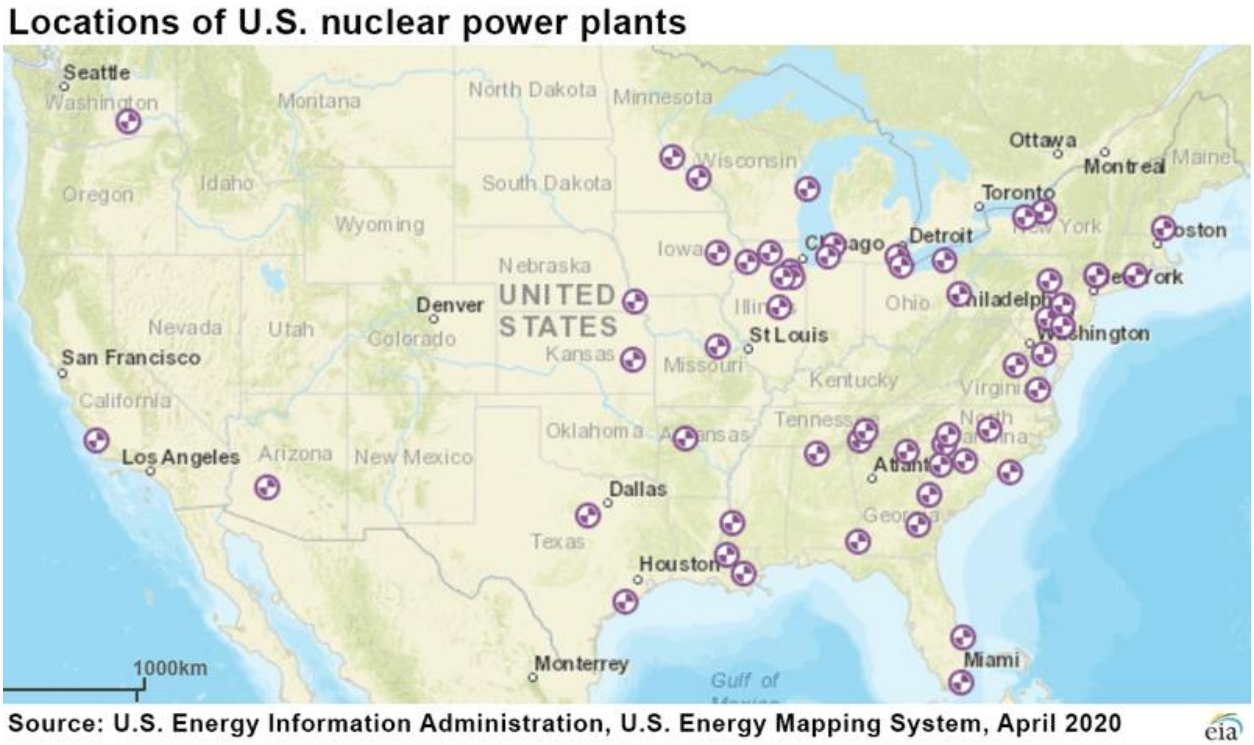


NUCLEAR POWER'S GREAT POTENTIAL: PERMANENTLY ELIMINATING UNSUSTAINABLE SOURCES

By Kento Mencel Instructor: Catherine Winiarski June 3rd, 2021

The United States has historically been dependent on fossil fuels, and it has only worsened over time; As shown in figure 1 below, while coal has decreased, our overall dependance on unsustainable resources has increased exponentially in the last decade. Most of our electricity is generated from unsustainable sources of energy. While there has been an increase in the use of nuclear and renewables since the 1970's, they roughly only account for 20% of our total energy production, far below the range that would reverse the continuous ravaging effects of climate change and greenhouse gas emission in our atmosphere.

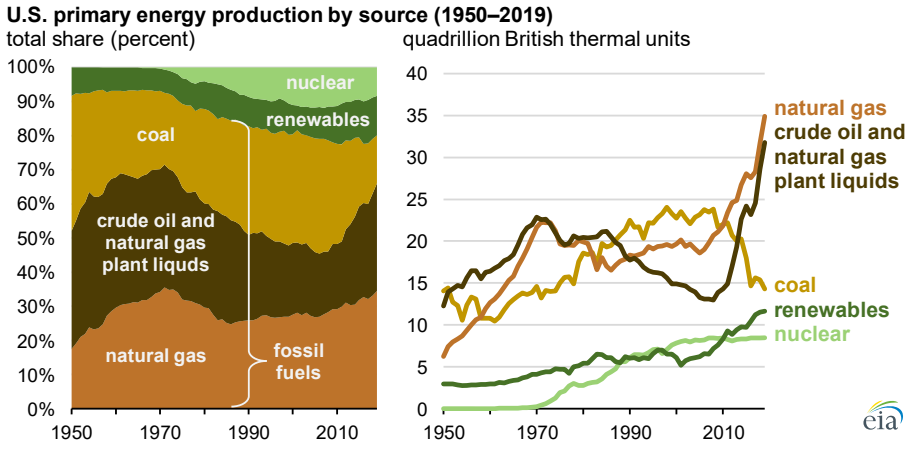


Figure 1: Breakdown of energy production from different sources, nuclear being around 10%. Source: eia.gov

NUCLEAR'S POTENTIAL

Nuclear energy is the key energy source available that will accelerate the transition away from fossil fuels. This is due to its positive qualities such as a decrease in waste, low operating costs, and a relatively low death toll compared to fossil fuels.

According to waste360.com, due to solar energy's limited shelf life, 6 million metric tons of e-waste is accumulated annually. Currently there is around 275,000 metric tons of spent nuclear waste stored globally. Additionally, as stated by whatisnuclear.com, Nuclear waste is also reusable, and you could power the entire US electricity grid off the energy in currently stored nuclear waste for almost 100 years. Additionally, in regards to the nuclear energy plants that are in operation in the United States, nuclear waste has been able to be safely contained and managed due to correct practices, as stated by the U.S. Energy Information Administration.

This demonstrates that nuclear power operates with relatively low overhead, and it has more expandability than other sources such as solar and wind as its core materials can eventually be reused. Additionally, nuclear power plants require a low amount of physical material to produce and does not require as much land as solar or wind. 40 metric tons of metals are necessary to facilitate the construction of modern power plants, which produce an average of 1 gigawatt of power.
http://www.nextbigfuture.com

According to Energy.gov, this amount of power is equivalent to 3.125 million PV solar panels or 431 utility scale wind turbines. Energy.gov This data shows that nuclear energy takes much less material to produce and saves the planet tons of e-waste each year.

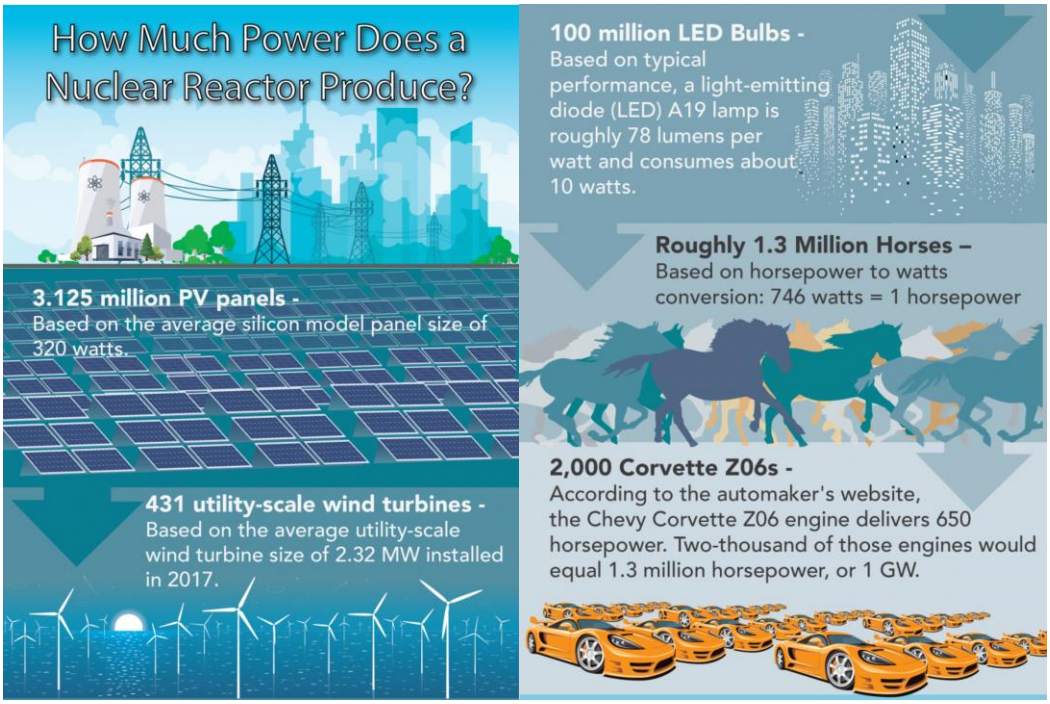


Figure 2: Equivalent power output of one nuclear reactor to other forms of energy. Source: energy.gov

From the mere 94 nuclear energy plants that are operating in the United States, they account for 20% of our electricity generation. According to Energy.gov, "They also provide more clean energy to the grid than any other energy source, accounting for more than half of the country's clean energy electricity production." Energy.gov

Modern nuclear power plants achieve the goal that climate scientists are fighting to achieve: the reduction of greenhouse gas, replacement of unsustainable energy sources, a quick transition away from fossil fuels, and a safer alternative.

Nuclear energy is relatively safe when we consider it as a replacement for fossil fuels such as brown coal, coal, oil, and natural gas. Examining the death rates from energy production per terawatt-hour, nuclear energy's death rate is 0.07, compared to 32.72 for brown coal, 24.62 for coal, and 18.43 for oil (OurWorldInData.org/energy). As a fossil fuel replacement, nuclear energy is highly viable as a source of electricity due to the lesser threat they pose to human lives when compared to the damage that results from unsustainable energy sources. Examining the data further, nuclear energy's death rate of 0.07 aligns more closely to wind power at 0.04, solar at 0.02, and hydropower at 0.02 (ourworldindata.org).

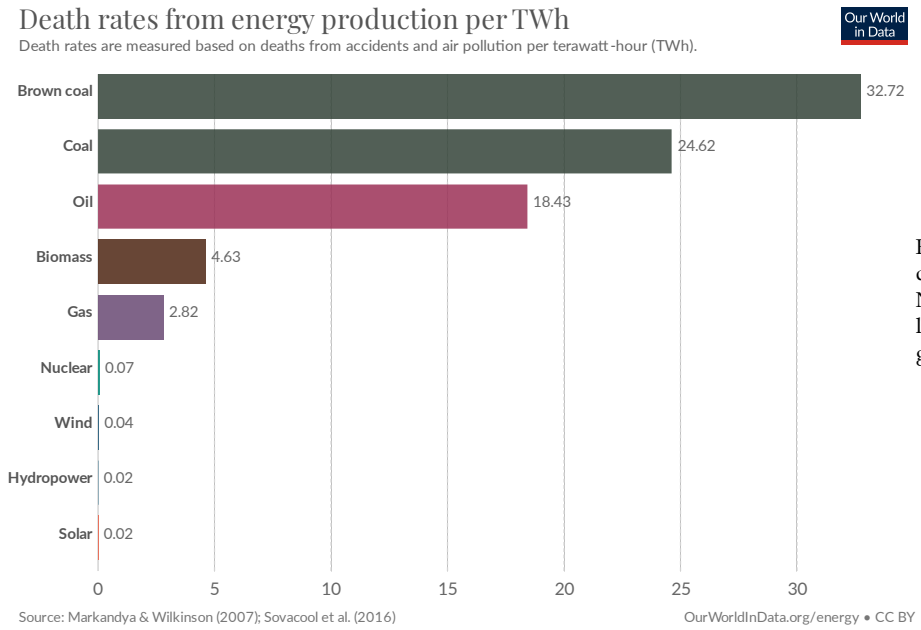


Figure 3: Death rates of different types of different types of energy, compared. Nuclear's rate is 0.07 per TWh, significantly lower than any fossil fuels, including natural gas. Source: ourworldindata.com

Events such as the 2011 Fukushima nuclear meltdown have distorted the public perception of nuclear energy. While nuclear meltdown situations are not ideal, zero deaths resulted in the radiation from the Fukushima meltdown (forbes.com). Additionally, the WHO stated that "for the general population inside and outside of Japan, the predicted risks are low and no observable increases in cancer rates above baseline rates are anticipated" (who.int). This was due to nuclear radiation spreading across the ocean and posed a small effect to human lives due to dilution.

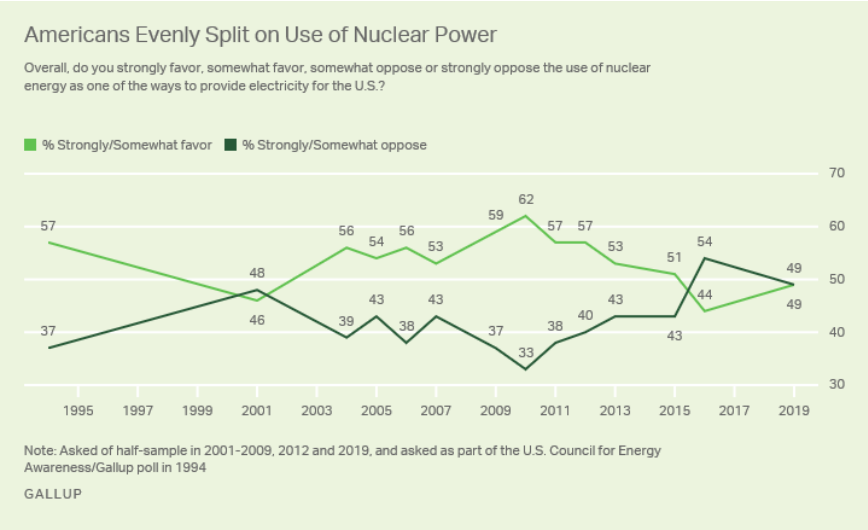
In the case of the 1986 Chernobyl meltdown, two hundred deaths were attributed to the radiation emitted from the failed plant. forbes.com Although these numbers seem large, the Soviet Union's management of the Chernobyl nuclear plant was different from nuclear plant designs we see today in the US, and the accident was attributed to a flawed reactor design as well as untrained workers. According to the NRC, "Many factors protect U.S. reactors against the combination of lapses that led to the accident at Chernobyl. Differences in plant design, broader safe shutdown capabilities and strong structures to hold in radioactive materials all help ensure U.S. reactors can keep the public safe." nrc.gov While attempting not to disregard the tragic death toll that occurred in Chernobyl due to the meltdown, the cause of the meltdown was due to errors in their construction and operation, not due to nuclear energy's attributes. These deaths, while still tragic, were avoidable, and the Soviet Union's neglect holds more responsibility for these deaths than the act of operating a nuclear plant, and incorrect practices were used as well as a under designed plant. nrc.gov

While events such as WWII, Fukushima, and Chernobyl are important to understand the dangers behind nuclear energy, the modern operating standards of nuclear plants, a relatively low death toll from nuclear related incidents, and the relatively minor effects of nuclear plant radiation on the environment demonstrates nuclear energy's best use case – to replace fossil fuels at the fastest rate.

SOCIAL MISCONCEPTIONS

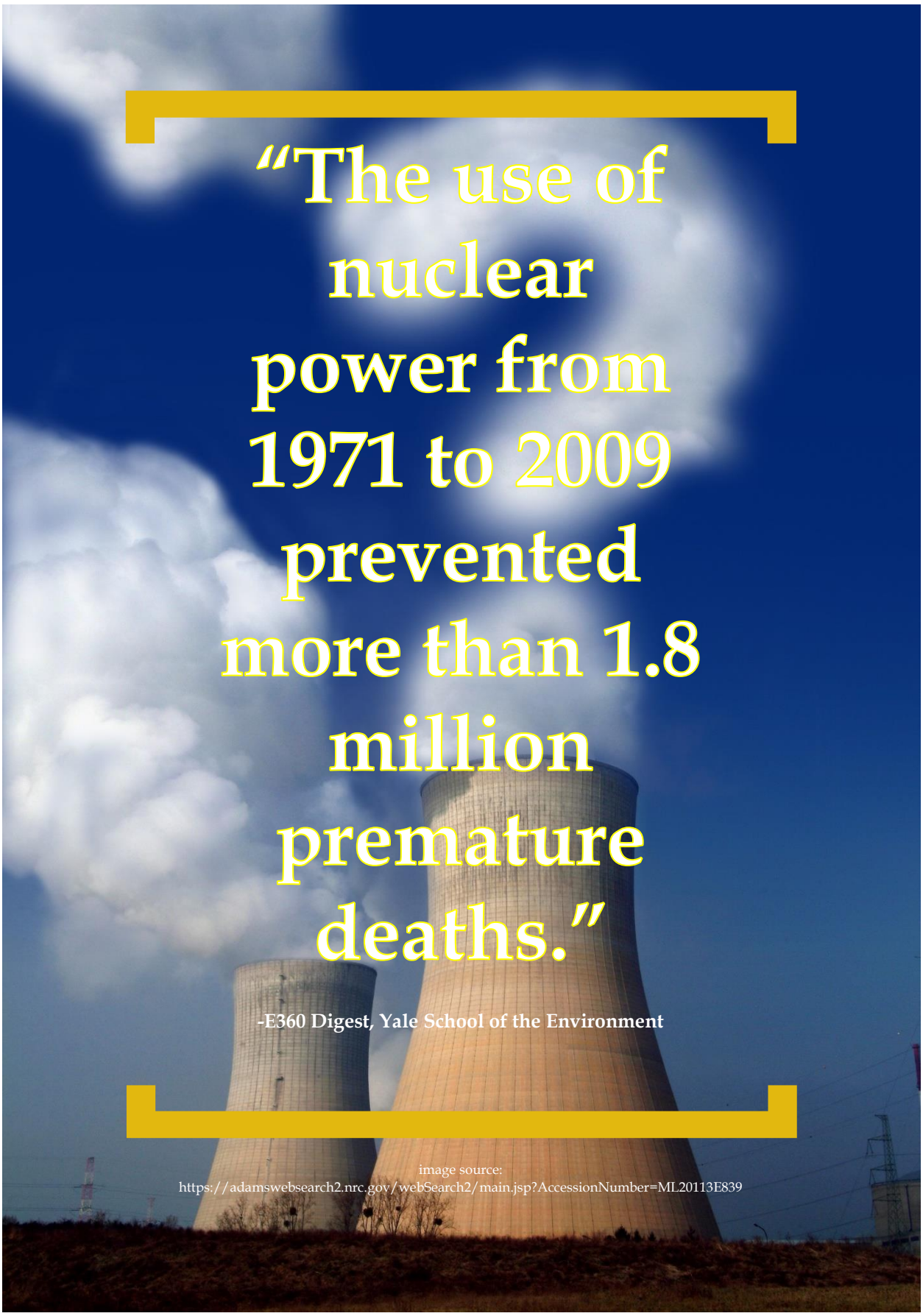
A study published by NASA scientists Pushker A. Kharecha and James Hansen suggested that 1.8 million air pollution-related deaths were prevented by nuclear energy between 1971-2009. They also state that by 2050, up to an additional 7 million lives could be saved (pubs.acs.org). This is mainly due to the energy source it replaced over the decades, which is fossil fuels. Through nuclear, our overall carbon footprint can be reduced without crippling the high energy needs in the United States at a high rate of speed and efficiency compared to other renewables. While common to attribute the idea of nuclear energy with meltdowns, death, and destruction, nuclear energy has been silently saving millions of lives for years.

The safety of generating nuclear energy has been a highly debated topic for the last few decades, highly attributable to nuclear meltdowns such as Fukushima and Chernobyl, the invention and deadly use of U.S. nuclear warheads codenamed "Fat Man" (Plutonium) and "Little Boy" (Uranium), and the nearly 5 decade long Cold War. With so much deadly history and misinformation surrounding the use of nuclear energy, public opinion of nuclear-based generators has been split down the middle in the US, as according to a Gallup poll, 49 percent are in favor of nuclear energy and 49 percent are opposed. From a political standpoint, 65% of Republicans support nuclear, while only 43% of Democrats support nuclear (news.gallup.com).



Poll on acceptance of nuclear energy in the United States. Source: news.gallup.com

Nuclear energy's accelerated plant construction needs bipartisan support, and I urge you to contact your local governmental representatives and voice your support for nuclear energy. Changing the public perception on nuclear while convincing our legislators to lobby against the use of fossil fuels and increase spending on nuclear is the only time effective way to ensure the viability of our future generations and prevent air-pollution related deaths at a high rate. While sources such as wind and solar are helpful in this process, their downsides and high costs hold them back to the potential energy output at low costs that nuclear provides.


"The use of nuclear power from 1971 to 2009 prevented more than 1.8 million premature deaths."
-E360 Digest, Yale School of the Environment
image source:
https://adamswebsearch2.nrc.gov/webSearch2/main.jsp?AccessionNumber=ML20113E839